Day 2 Student Notes: Tokens, Context Window and Pricing

Claude for DevOps Mastery. Keep these notes: you will use the formula and the checklist all through the course.

> **Prices change.** The numbers below were reported in October 2026. Before you pay for anything, check claude.com/pricing and the Claude API documentation.

What we covered

A token is a small chunk of text. Claude reads tokens, not letters or words. In English, 1 token is about three-quarters of a word, so **tokens = words x 1.3**. Examples: 3 words is about 4 tokens, 100 words is about 130 tokens, one page of 500 words is about 650 tokens.

Input tokens are everything you send: your question, pasted text, files and earlier messages. **Output tokens** are everything Claude writes back. On the API, output costs more than input. Asking for "3 bullets" or "under 80 words" saves money and time.

The context window is Claude's notebook. Your messages and Claude's replies share the same pages. Many models offer about 200K tokens (roughly 150,000 words), some offer 1M. A new chat is a blank notebook. If a chat gets very long, ask Claude for a short summary and paste it into a new chat.

Key words

Word	Meaning
Token	A small chunk of text, about 3/4 of a word
Input tokens	Everything you send to the model
Output tokens	Everything the model writes back
Context window	How much Claude can use in one conversation, input and output together
Plan	A monthly subscription for people (Free, Pro, Max, Team, Enterprise)
API	Pay-per-token access for developers building their own tools
Prompt caching	Repeated text is read at about 10% of the input price
Batch API	About 50% cheaper for jobs that can wait

Claude plans (reported October 2026)

Plan	Price in USD	Good to know
Free	$0	Daily limits. Enough for Days 1 to 3
Pro	$20 a month (about $17 billed yearly)	More usage, Projects, Claude Code, connectors
Max	$100 or $200 a month	Heavy daily use. Not needed for this course
Team	From $25 per seat a month	Shared workspace, admin controls
Enterprise	Custom	SSO, audit logs, security controls

For this course: Free for Days 1 to 3. **Claude Pro from Day 4. Subscribe on the evening of Day 3.** You do not need Max or the API. Pro limits are shared between chat and Claude Code.

API prices per 1 million tokens (reported October 2026)

Model	Input	Output
Haiku 5.5	$0.10	$0.50
Sonnet 5.5	$2	$10
Opus 5.5	$4	$20
Fable 5.1 (Mythos tier)	Premium. See the pricing page	

The cost formula

> **cost = (input tokens x input price + output tokens x output price) / 1,000,000**

Example on Sonnet 5.5: 1,000 tokens in and 500 tokens out. Input: 1,000 x $2 / 1,000,000 = $0.002. Output: 500 x $10 / 1,000,000 = $0.005. Total $0.007, less than 1 cent.

Practical C: price an email task

Paste the sample email and prompt from the prompts file. Count the words in Claude's answer and multiply by 1.3. With about 150 tokens in and 150 tokens out:

Model	Input cost	Output cost	One email	1,000 emails
Haiku 5.5	$0.000015	$0.000075	$0.00009	about $0.09
Sonnet 5.5	$0.0003	$0.0015	$0.0018	about $1.80
Opus 5.5	$0.0006	$0.003	$0.0036	about $3.60

Notice: one email costs a fraction of a cent, so cost matters at volume. Output costs more than input, so ask for short answers. Cheaper is not always right: compare what each model wrote.

Your own table: fill in your numbers here and compare.

Model	Input tokens	Output tokens	Cost
Haiku 5.5			
Sonnet 5.5			
Opus 5.5			

Which model should I start with?

Task	Start with	Why
Fix grammar, summarise an email, explain an error	Haiku 5.5	Fast and cheapest
Write a report, a script, a config; most daily work	Sonnet 5.5	Best balance. Our default
Hard analysis, tricky bug, architecture decision	Opus 5.5	Deepest reasoning
Specialist or select enterprise work	Fable 5.1 (Mythos tier)	Rarely needed for us

Rule: start small. If the answer is wrong or shallow, go up one tier. Whatever the model, verify before you use it.

Check yourself

1. About how many tokens is a 100-word message?
2. What takes up space in the context window?
3. What happens when you open a new chat?
4. Which costs more per token on the API, input or output?
5. Which plan do we need from Day 4, and when do we subscribe?

Answers

1. About 130 tokens (words x 1.3).
2. Your messages and Claude's replies, together.
3. It starts with a blank notebook and does not know the earlier chat.
4. Output (about 5 times more in the table).
5. Claude Pro, from Day 4. Subscribe on the evening of Day 3.

Take-home task (about 20 minutes)

1. Pick one real task from your week: an email, a summary, a script you want explained.
2. Estimate input words, expected output words, convert to tokens, and work out the cost on Haiku, Sonnet and Opus.
3. Write down 3 tasks you do often. Bring them to Day 3.
4. Do not paste passwords, secrets or company data into any chat.
5. Share one number or one surprise in the batch group.

Day 3 preview: prompt engineering. Role, context, constraints, format and examples.
Reminder: subscribe to Claude Pro on the evening of Day 3. Check the current price on the official pricing page first.